Garbage Classification Using Deep Learning

MobileNetV2 Transfer Learning Approach

Introduction

- Waste management is one of the most critical environmental challenges in modern society
- Proper classification of garbage plays a major role in recycling and reducing pollution
- Manual sorting is time-consuming, expensive, and prone to human error
- Automating the process using AI and Deep Learning is an effective solution
- This project builds an intelligent image classification system to automatically categorize waste

Abstract

- Deep learning-based garbage classification system using Garbage Classification V2 dataset (Kaggle)
- Dataset contains 10 categories: plastic, glass, metal, paper, clothes, biological waste, etc.
- MobileNetV2 pretrained model used as feature extractor with custom classification layers
- Trained using TensorFlow and Keras with data augmentation and normalization
- Achieves ~92% validation accuracy with strong generalization across waste categories

Problem Definition

- Traditional waste sorting methods suffer from several limitations:
- • High dependency on manual labor
- • Human errors in classification
- • Inefficiency in large-scale recycling systems
- • Increased environmental and operational costs

- Main Problem: How can we automatically and accurately classify different types of garbage images into predefined categories using deep learning?

Solution Overview

- Developed a deep learning pipeline with the following approach:
- • Dataset: Garbage Classification V2 from Kaggle (~19,000+ images, 10 classes)
- • Data Preprocessing: 224×224 resize, pixel normalization, 80/20 train-validation split
- • Model: MobileNetV2 (ImageNet pretrained) with transfer learning
- • Added custom layers: Global Average Pooling, Dense (128), Dropout, Softmax output
- • Training: Adam optimizer, Categorical Crossentropy, EarlyStopping & ModelCheckpoint
- • Evaluation: Accuracy, Loss, and Confusion Matrix analysis

Dataset & Data Preprocessing

Dataset Details

- Source: Garbage Classification V2 (Kaggle)
- Total Images: ~19,000+
- Classes: 10 categories
- Categories: Plastic, Glass, Metal, Paper, Clothes, Biological, etc.

Preprocessing Steps

- Images resized to 224×224 pixels
- Pixel normalization: rescale = 1/255
- Data split: 80% Training, 20% Validation
- ImageDataGenerator for augmentation & normalization

Model Architecture

- Base Model: MobileNetV2 (pretrained on ImageNet)
- Transfer Learning: Base model frozen during initial training
- Added Custom Layers:
- • Global Average Pooling layer
- • Dense layer (128 neurons, ReLU activation)
- • Dropout layers (0.4, 0.3) for regularization
- • Output layer with Softmax (10 classes)

Training Strategy

Hyperparameters

- Optimizer: Adam
- Loss Function: Categorical Crossentropy
- Batch Size: 32
- Epochs: 5

Callbacks & Techniques

- EarlyStopping: Prevents overfitting
- ModelCheckpoint: Saves best model
- Dropout regularization (0.4, 0.3)
- Data augmentation via ImageDataGenerator

Evaluation & Metrics

- Model evaluated using validation dataset
- Key Metrics: Accuracy and Loss
- Confusion Matrix for detailed error analysis
- Visual Outputs Generated:
- • Training vs Validation Accuracy graph
- • Training vs Validation Loss graph
- • Sample image predictions (True vs Predicted labels)
- • Confusion Matrix heatmap

Results & Performance

Validation Accuracy

~92.68%

Strong performance across all waste categories

Classification

10 Categories

Effective multi-class waste identification

Overfitting Control

Low

Dropout and EarlyStopping prevent overfitting

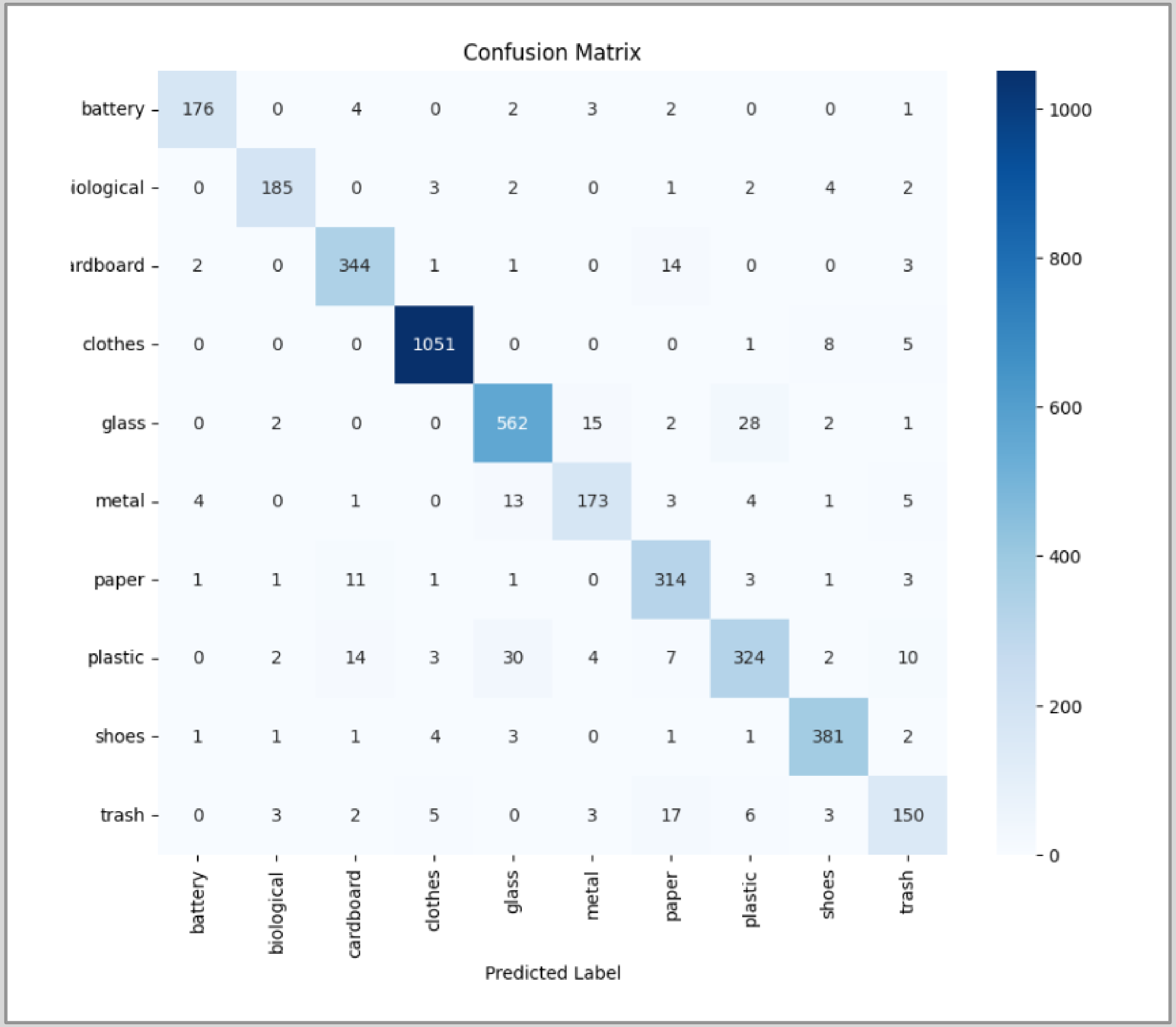
Generalization

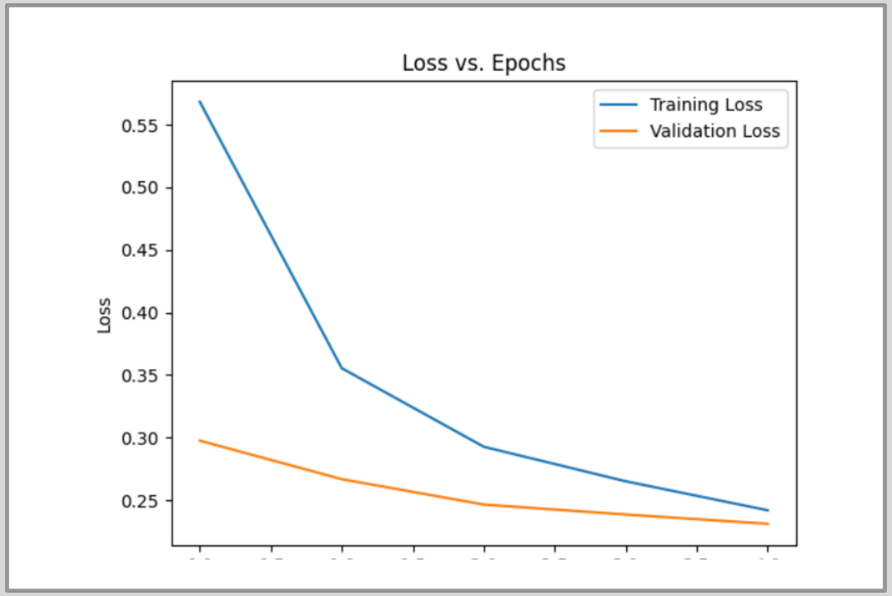
Strong

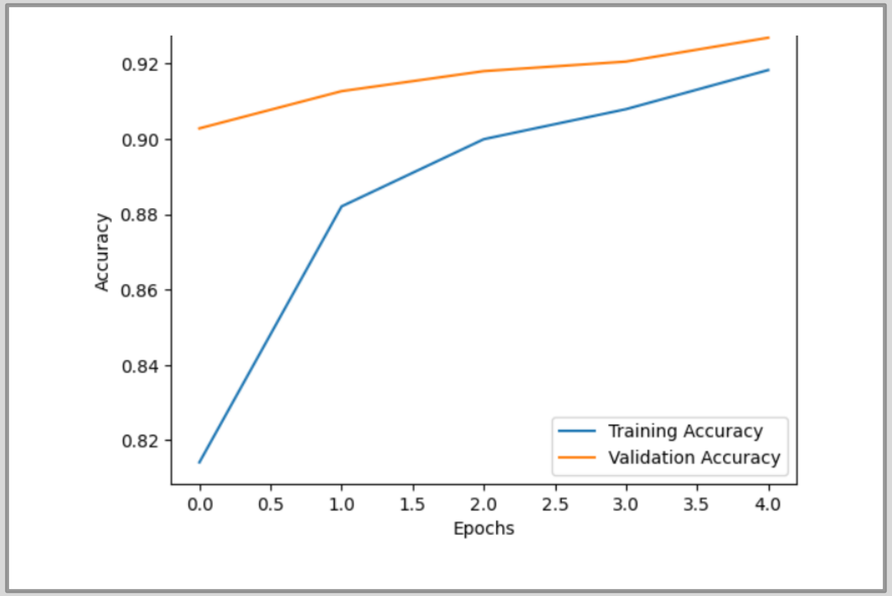
Model performs well on unseen validation data

Visual Outputs Generated

- Training vs Validation Accuracy Graph — shows model convergence
- Training vs Validation Loss Graph — monitors learning stability
- Sample Image Predictions — True vs Predicted labels comparison
- Confusion Matrix Heatmap — detailed per-class performance analysis
- All visualizations confirm effective learning and strong generalization


Confusion Matrix
battery 176 0 4 0 2 3 2 0 0 1
iological 0 185 0 3 2 0 1 2 4 2
ardboard 2 0 344 1 1 0 14 0 0 3
clothes 0 0 0 1051 0 0 0 1 8 5
glass 0 2 0 0 562 15 2 28 2 1
metal 4 0 1 0 13 173 3 4 1 5
paper 1 1 11 1 1 0 314 3 1 3
plastic 0 2 14 3 30 4 7 324 2 10
shoes 1 1 1 4 3 0 1 1 381 2
trash 0 3 2 5 0 3 17 6 3 150
battery biological cardboard clothes glass metal paper plastic shoes trash
Predicted Label
1000
800
600
400
200
0


Loss vs. Epochs
Training Loss
Validation Loss
Loss
0.55
0.50
0.45
0.40
0.35
0.30
0.25


Accuracy
0.92
0.90
0.88
0.86
0.84
0.82
Training Accuracy
Validation Accuracy
0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0
Epochs

True: glass | Pred: glass

True: shoes | Pred: shoes

Conclusion

- The model successfully distinguishes between different types of waste with high accuracy
- Transfer learning using MobileNetV2 proves highly effective for garbage classification
- The system demonstrates strong generalization across 10 waste categories
- Low overfitting achieved through dropout layers and early stopping callbacks
- This approach can be scaled for real-world automated waste sorting applications
- Future work: Expand dataset, fine-tune base model, deploy to edge devices

Thank You

Garbage Classification Using Deep Learning (MobileNetV2)